

BPhO ROUND 0 · QUESTION PAPER

BPhO Round 0 — 2025 paper

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

The one real Round 0 paper in existence. Reproduced from the published BPhO material for personal study.

Before you start

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · Work done stretching a wire E

A uniform elastic wire of length L and cross-section A is subject to a tensile stress σ , which results in a tensile strain ϵ . Find the work done on the wire.

A $LA\sigma\epsilon$

B $\sigma A / (\epsilon L)$

C $\sigma L / (2\epsilon A)$

D $\epsilon A / (\sigma L)$

E $\frac{1}{2}LA\sigma\epsilon$

Q5 · Interference at a given path difference **F**

Two coherent waves of equal amplitude and wavelength 1.8 m, originating from the same source, arrive at the same detector with a path difference of 7.5 m. Which of these options describes the type of interference that occurs at the detector?

- A** fully destructive
 - B** mostly destructive
 - C** mostly constructive
 - D** fully constructive
 - E** not enough information
-

Q6 · Cheapest network of exactly 4 Ω **H**

A resistor shop sells individual resistors (the stock is unlimited). The price for each resistor is as follows: 2 Ω for £3, 3 Ω for £1, 6 Ω for £3, 8 Ω for £4, 12 Ω for £2.

What is the least you would need to spend to make a combination of resistors equivalent to exactly 4 Ω ?

- A** £3
 - B** £4
 - C** £5
 - D** £6
 - E** £7
-

Q7 · Relative velocity: catching up **B**

Two particles A & B travel along a straight line at a constant speed with velocities as shown in the diagram. Their initial separation is 18 m. How far does particle A travel before colliding with B?

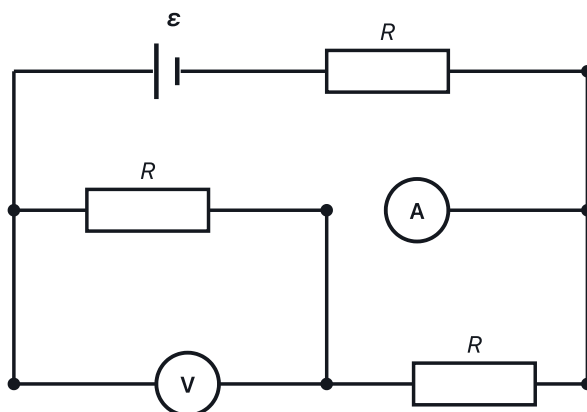


Read the arrows before you reach for a formula. The two velocity arrows point the same way. That single observation decides which arithmetic applies — and the diagram is the only place it is stated.

- A** 18 m
 - B** 27 m
 - C** 36 m
 - D** 45 m
 - E** 54 m
-

Q8 · Ideal voltmeter and ammeter readings H

A cell of emf ε (and negligible internal resistance) is connected to an ideal voltmeter, an ideal ammeter and three resistors R as shown below. What are the readings on the voltmeter and ammeter respectively?



The circuit as printed. Five components, two of them meters. Before you write anything down, decide what an *ideal* ammeter and an *ideal* voltmeter each do to the branch they sit in.

- | | |
|---|---|
| A $\varepsilon/3$, $\varepsilon/(3R)$ | B $\varepsilon/3$, $\varepsilon/(2R)$ |
| C $\varepsilon/3$, ε/R | D $\varepsilon/2$, $\varepsilon/(3R)$ |
| E $\varepsilon/2$, $\varepsilon/(2R)$ | |

Q9 · First-harmonic frequency against resistance F

Two uniform metal wires of the same length are made from the same material. Separately, the wires are subject to the same force on both ends, and the first harmonic frequency of each is measured. The frequencies are in the ratio 2 : 1. What is the ratio of their electrical resistances?

[Hint: the speed of a transverse wave on a string is $v = \sqrt{(T/\mu)}$.]

- | | |
|-------------------------|----------------|
| A 4 : 1 | B 3 : 1 |
| C $\sqrt{3}$: 1 | D 2 : 1 |
| E $\sqrt{2}$: 1 | |

Q10 · Estimating the number of atoms in the Earth A

Estimate the number of atoms which make up the Earth.

- | | |
|--------------------|--------------------|
| A 10^{50} | B 10^{55} |
| C 10^{60} | D 10^{65} |
| E 10^{70} | |

Q15 · How many ice cubes the water can melt J

An insulated cup contains 500 g of water at 20 °C. Identical ice cubes, each of mass 25 g and at 0 °C, are added to the cup. What is the maximum number of cubes that can be added so that, at equilibrium, the mixture is entirely liquid?

Specific heat capacity of water = $4 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice = $3 \times 10^5 \text{ J kg}^{-1}$

- | | |
|------------|------------|
| A 4 | B 5 |
| C 6 | D 7 |
| E 8 | |
-

Q16 · Time of flight through a glass cube G

A ray of light is incident on a glass cube of side length a and refractive index $\sqrt{2}$. The angle of incidence is 30°. Find the time taken for the light to travel to the opposite face.

- | | |
|-----------------------------------|-----------------------------------|
| A $\sqrt{8/7} \cdot a/c$ | B $(4/\sqrt{7}) \cdot a/c$ |
| C $(\sqrt{2/3}) \cdot a/c$ | D $(4/\sqrt{3}) \cdot a/c$ |
| E $(4/\sqrt{2}) \cdot a/c$ | |
-

Q17 · How many photoelectrons charge the capacitor L

An uncharged, isolated parallel plate capacitor has capacitance C . Monochromatic light of frequency f is incident on one of the metal plates, which has threshold frequency f_0 . Emitted photoelectrons are then captured by the other plate. Once the capacitor charge has reached a steady value, how many electrons have been transferred between the plates? (h is Planck's constant and e is the magnitude of the electron charge.)

- | | |
|----------------------------|------------------------------|
| A $Ch(f + f_0)/e$ | B $Cf_0(f + f_0)/e^2$ |
| C $Ch(f - f_0)/e$ | D $Cf(f + f_0)/e^2$ |
| E $Ch(f - f_0)/e^2$ | |
-

Q18 · Ranges at complementary angles B

Two particles are projected from ground level at the same speed v , but at two different angles to the horizontal: $(\pi/4 + \alpha)$ and $(\pi/4 - \alpha)$, where α itself is a small angle in radians. Find the difference between the horizontal ranges of the two particles.

- | | |
|--------------------------|----------------------------|
| A 0 | B $\alpha v^2/g$ |
| C $2\alpha v^2/g$ | D $\alpha v^2/(2g)$ |
| E $4\alpha v^2/g$ | |
-

Q19 · Time for one third to decay K

The half-life of a radioactive substance is T . What is the time taken for one third of the substance to decay?

A $(2/3) T$

B $\sqrt{(2/3)} \cdot T$

C $T \log_2(3/2)$

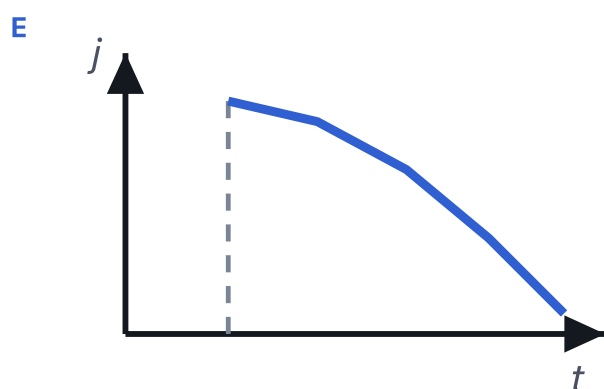
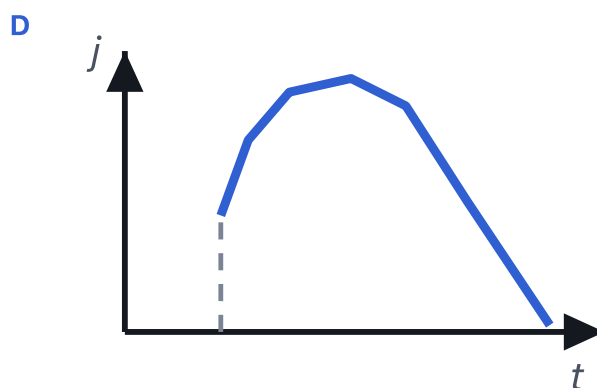
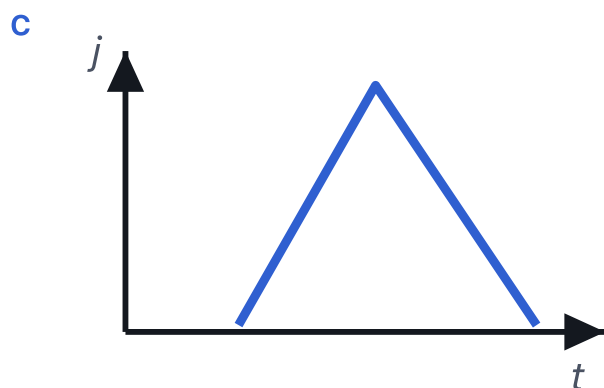
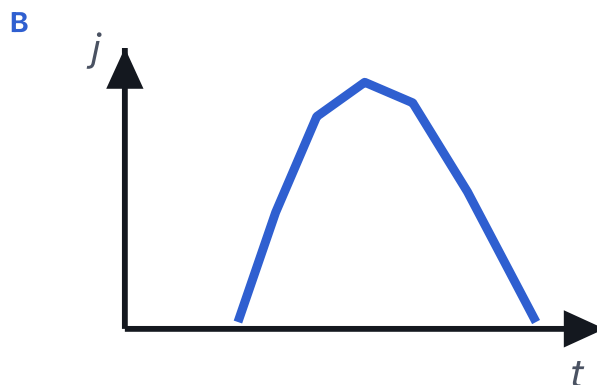
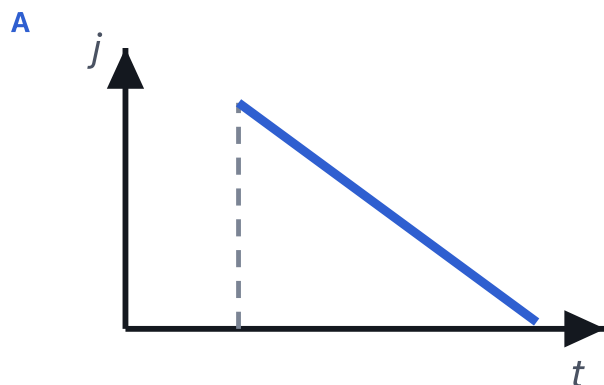
D $T \log_2(3)$

E $T^{2/3}$

Q20 · Jerk of a bungee jumper against time C

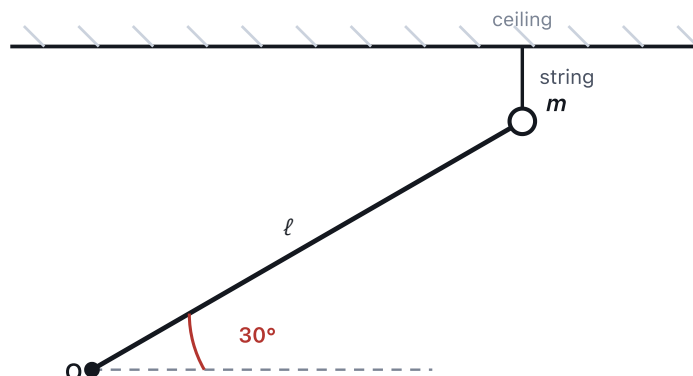
Jerk is defined as the rate of change of acceleration with respect to time, $j = da/dt$.

A bungee jumper attached to a slack rope falls from a height. Once the rope is taut, it acts as an ideal spring which subsequently brings the jumper to instantaneous rest. Which of the following graphs shows the jerk j as a function of time t , up to this moment?



Q21 · Moments C

A light inextensible string is attached to one end of a rigid rod of negligible weight and of length ℓ , with its other end freely hinged at O. The other end of the string is attached to the ceiling, so that the string is vertical, and the rod makes an angle of 30° to the horizontal. A small smooth ring of mass m is released from rest at the upper end of the rod and slides downwards along it.



The rod is held still by the string; the ring slides along it. Note which end the ring starts at, and that the rod keeps the ring at a fixed angle — the motion is one-dimensional along the rod, not vertical.

Given that the tension in the string T at time t (while the ring is in motion) is $T = mg(1 - kt^2)$, select the correct expression for k .

- | | |
|---------------------------------|--------------------------------|
| A $g / 4\ell$ | B $g / 2\ell$ |
| C g / ℓ | D $g / (\sqrt{2} \ell)$ |
| E $\sqrt{3} g / (2\ell)$ | |

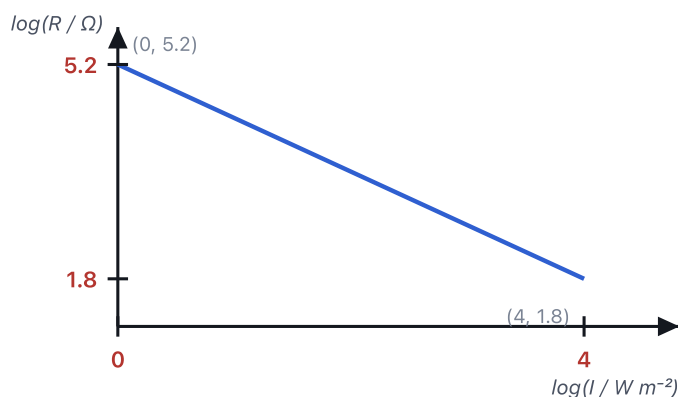
Q22 · Conical pendulum D

A conical pendulum consists of a bob attached to a light inextensible string of length ℓ , with the upper end fixed. The bob moves in a horizontal circle so that the string makes an angle of 30° to the vertical. By what factor must the angular frequency be increased for this angle to double to 60° ?

- | | |
|----------------------------|---------------------|
| A $\sqrt{\sqrt{3}}$ | B $\sqrt{2}$ |
| C $\sqrt{3}$ | D 2 |
| E 3 | |

Q23 · Log-log graphs H

The resistance R of a light-dependent resistor (LDR) depends on the light intensity I . A logarithmic plot relating these variables is shown below.



A straight line on log-log axes, so $R \propto I^n$ with n equal to the gradient. Both scales are logarithmic, and the two end points are marked. (Reading the gradient off the plot is part of the question; what you then do with a fractional power is the rest of it.)

A point light source placed a distance d from the LDR illuminates it, and its resistance is measured to be R_0 . Which of the following is a good approximation for the LDR resistance when the distance to the light source is halved?

- | | |
|--------------------|--------------------|
| A $0.1 R_0$ | B $0.3 R_0$ |
| C $0.5 R_0$ | D $0.7 R_0$ |
| E $0.9 R_0$ | |

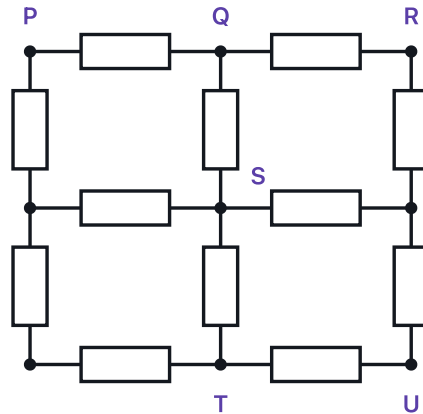
Q24 · Centre of mass C

A uniform ladder of mass m rests against a wall in static equilibrium at an angle of 45° to the vertical. The base of the ladder is pushed a small distance a towards the wall. Which of these options gives the best approximation for the change in gravitational potential energy of the ladder?

- | | |
|----------------------------|-----------------------------|
| A $\frac{1}{2} mga$ | B $(1/\sqrt{2}) mga$ |
| C mga | D $\sqrt{2} mga$ |
| E $2 mga$ | |

Q25 · Network reduction H

Consider the arrangement of 12 identical resistors below. The equivalent resistance is measured between the following pairs of points: PR, PS, PU, QS, QT. Which measurement gives the median (middle value) resistance?



A 3×3 grid of nodes with all twelve resistors identical. The labels sit on the nodes: P, Q, R along the top; S at the centre; T and U along the bottom. Every line you can see is one resistor.

- | | |
|-------------|-------------|
| A PR | B PS |
| C PU | D QS |
| E QT | |